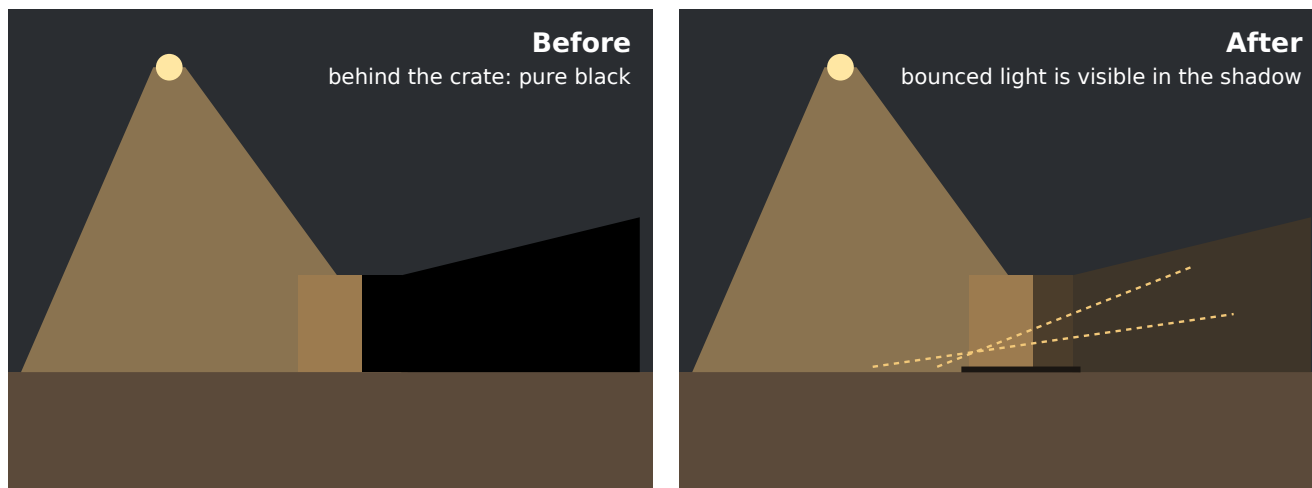


No more pitch-black shadows in Quake 4

How to install **openQ4 Light**: soft bounce light in shadows and proper ambient occlusion (AO). No programming - just copy some files and click a couple of times.



In the original Quake 4 everything a lamp does not reach is completely black. This pack adds the light that in real life bounces off walls and floors, and proper shading in corners and gaps.

What you need

	What	Notes
✓	Quake 4 from Steam or GOG	A regular purchased copy of the game
✓	openQ4 0.13.2 or newer	Open-source engine for Quake 4, github.com/themuffinator/openQ4
✓	A GPU with OpenGL 4.5	Anything released since about 2014
✓	Disk space: ~130 GB free on any drive	For a temporary paging file while baking
✓	One night	Baking the whole campaign takes about 10 hours; you don't need to touch the PC meanwhile

The whole process - 4 steps

Step	What we do	Time
1	Enlarge the Windows paging file	3 minutes + reboot
2	Bake the light for the maps	one night
3	Put the preset file into the game folder	1 minute
4	Type one command in the game console	1 minute

What goes where

First, the two folders: the pack folder (the one you downloaded) and the game folder.

Pack folder

openQ4_Light	
GUIDE_EN.pdf	this guide
README.md	the same, short
1_PRESETS	needed: pick one preset
2_BAKE	needed: files for baking
3_SINGLE_SHADERS	for experiments, not needed
4_SOURCE	sources and licence, not needed
5_DEV	for developers, not needed

A regular player only needs **1_PRESETS** and **2_BAKE**. You can ignore the rest.

Game folder

This is the folder with **openQ4-client_x64.exe**. Next to it there is a **baseoq4** folder - presets go there.

Quake 4 (or openQ4)	
baseoq4	here - the preset file (step 3)
pak0.pk4	openQ4 files, don't touch
pak1.pk4	
q4base	game files, don't touch
openQ4-client_x64.exe	here - files from 2_BAKE (step 2)

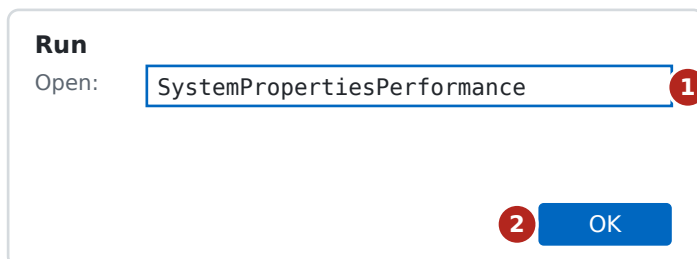
Tip. How to find the game folder: right-click the openQ4 shortcut -> "Open file location". If you used the openQ4 installer, it is usually %LOCALAPPDATA%\Programs\openQ4 (paste this path into the Explorer address bar). If you unpacked the archive into the Steam Quake 4 folder, it is the Quake 4 folder itself, e.g. ...\\steamapps\\common\\Quake 4.

Important. Do not edit or replace **pak0.pk4** and **pak1.pk4**: openQ4 checks them and will refuse to start. Our files start with **zz_**, the game loads them last, and they override what is needed on their own.

STEP 1

Enlarge the paging file

Why: in openQ4 0.13.2 memory use keeps growing during baking. With default Windows settings memory runs out and the bake crashes with **Out of memory**. A large paging file on disk solves this. Any drive works, even a regular HDD.

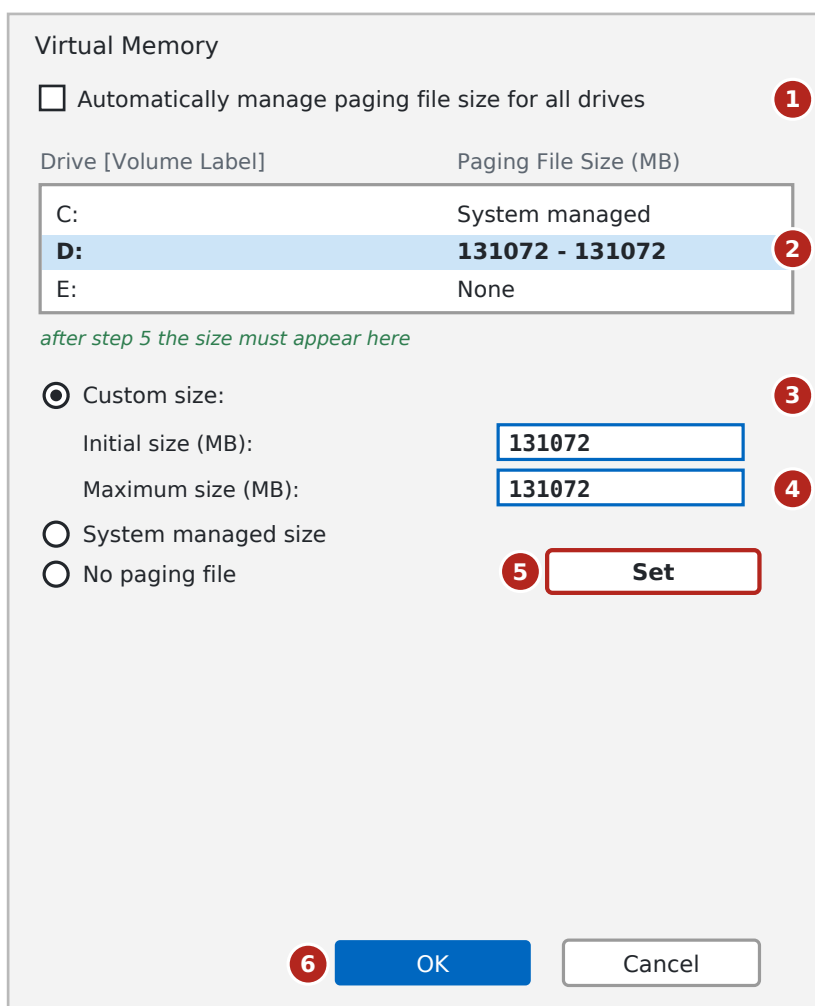


Run
Open: **1**
2 **OK**

Open the settings window.

Press **Win + R** (the Windows logo key and R), type SystemPropertiesPerformance **(1)** and press OK **(2)**.

In the window that opens, go to the **Advanced** tab and under Virtual memory click **Change....**



Virtual Memory

☐ Automatically manage paging file size for all drives **1**

Drive [Volume Label]	Paging File Size (MB)
C:	System managed
D:	131072 - 131072 2
E:	None

after step 5 the size must appear here

☒ Custom size: **3**
Initial size (MB):
Maximum size (MB): **4**
☐ System managed size
☐ No paging file

5 **Set**

6 **OK** Cancel

Fill in the Virtual Memory window:

1. Untick "Automatically manage paging file size for all drives".
2. Select a drive with plenty of free space in the list.
3. Choose "Custom size".
4. Enter the **same** initial and maximum size: **131072** and **131072**. If they differ, Windows doesn't grow the file in time and the bake fails.
5. Click **Set**. The selected drive in the list must now show **131072 - 131072**. If it still says "None", you did not click Set.
6. Click OK and restart the PC.

Tip. Once all maps are baked you can put everything back: tick "Automatically manage paging file size" again and restart.

STEP 2

Bake the light

"Baking" means calculating once, in advance, how much bounced light reaches every point of a map, and saving it to a file. openQ4 does the math; you only start it and wait.

2.1. Copy the files



All six files from **2_BAKE** go into the game folder, next to **openQ4-client_x64.exe**.

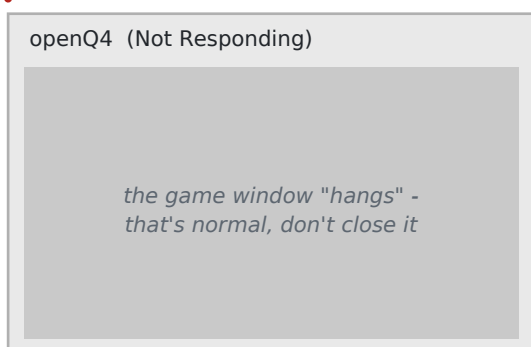
2.2. Pick one and run it

File	When to run	How long
bake_campaign_full.bat	For presets A and B . Whole campaign, full quality.	about 10 hours
<code>bake_campaign_quick.bat</code>	Only for preset C . Minimum quality.	quick
<code>bake_one_map.bat</code>	One map in full quality. Shows a list of maps to choose from.	15-25 minutes

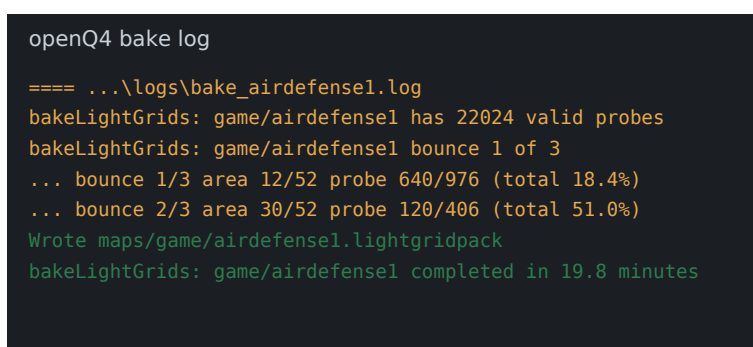
Double-click the file. If Windows asks whether to allow it, allow it.

2.3. What you will see

Important. Every map starts the game anew, and the game first sits on the **PRESS ANY KEY TO CONTINUE** screen. Until a key is pressed, the bake **does not start**. The `.bat` presses Space for you; if the screen stays up longer than 15 seconds, click the game window and press any key.



game window



bake progress window - watch this one

That's fine. The game window says "Not Responding" - that is expected; the black window shows the progress. When everything is done, the `.bat` window says "Done". You can stop at any time: on the next run finished maps are skipped. If a map runs out of memory (Out of memory), the `.bat` closes the game, moves on to the next map and at the end lists the maps that failed (also saved in `bake_failed.txt`).

STEP 3

Pick a preset

A preset is a single file that changes the lighting. All three are good, it's a matter of taste. If you are not sure, take **A**.

A Honest

Baked bounce light
AO darkens only that,
lamp light is untouched
Most physical look

File:
`zz_light_honest.pk4`

Console command:
`exec light_honest.cfg`

Needs: Full bake

B AO on whole frame

Baked light, but
never below a minimum
GTAO on the whole frame
More depth and contrast

File:
`zz_light_classic.pk4`

Console command:
`exec light_classic.cfg`

Needs: Full bake

C Simple

Flat shadow fill
no bounce light
GTAO on the whole frame
Fastest way

File:
`zz_light_simple.pk4`

Console command:
`exec light_simple.cfg`

Needs: Quick bake



Example for preset A. For B and C it's the same, with their own file.

Important. Keep **only one** preset in **baseoq4**. To switch presets, delete the old `zz_light_...` file and put in the new one.

STEP 4

One console command

Start the game as usual. Open the console with the `~` key; if it does not open, use **Ctrl + Alt + ~**. Type the command for your preset and press Enter:

```
Game console
exec light_honest.cfg      (preset A)
exec light_classic.cfg     (preset B)
exec light_simple.cfg      (preset C)
```

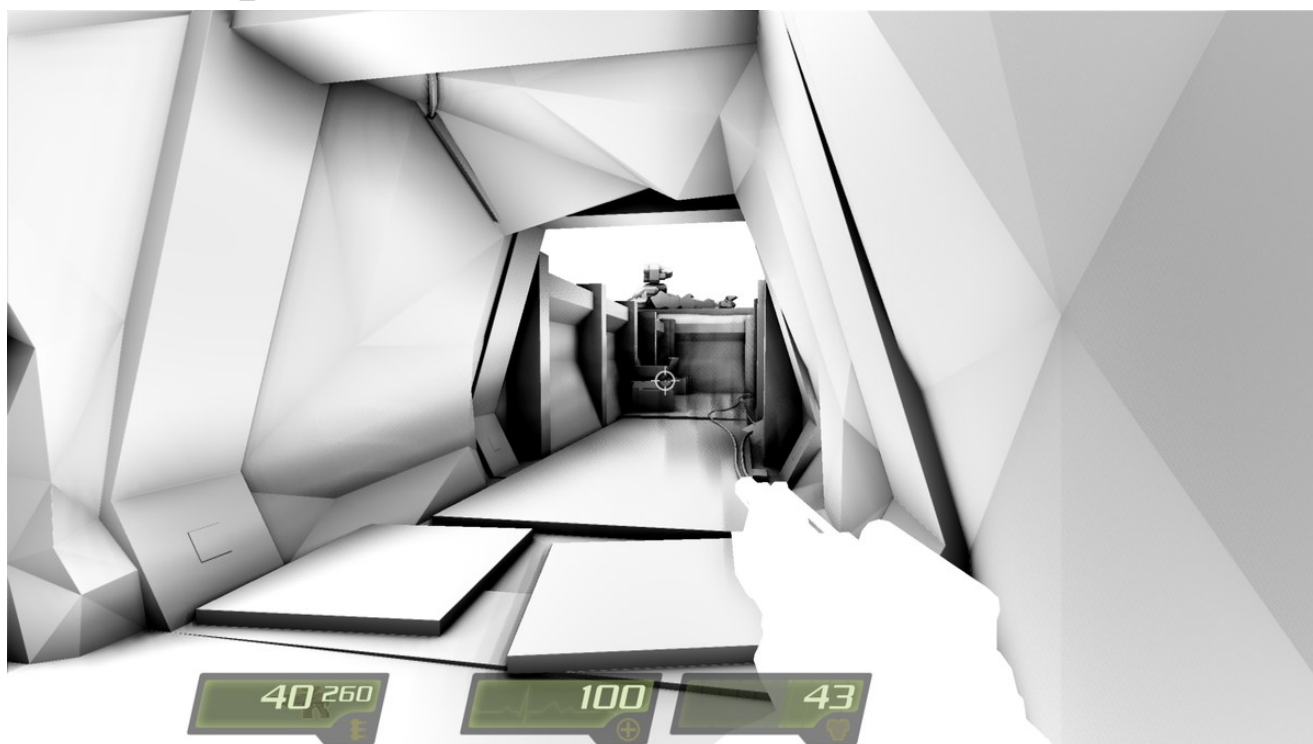
Type only **one** line - the one for your preset. The settings are saved; on later launches you don't need to type anything. That's it, go play.

How to check that it works

Type these in the game console. They break nothing and are turned off with the same command and 0.

Command	What it shows	Turn off
<code>r_lightGridDebug 2</code>	Only the baked bounce light, without textures. If the picture is black, the map is not baked.	<code>r_lightGridDebug 0</code>
<code>r_useLightGrid 0</code>	Turns the baked light off to compare before/after.	<code>r_useLightGrid 1</code>
<code>r_ssaoDebug 1</code>	Only the AO shading (presets B and C).	<code>r_ssaoDebug 0</code>
<code>path</code>	List of loaded archives. Your <code>zz_light_...</code> must be listed above <code>pak0.pk4</code> .	—

This is what `r_ssaoDebug 1` looks like



Dark: seams, corners, gaps and niches. Flat walls and floors stay light - as they should.

Why the new AO is better than the stock one



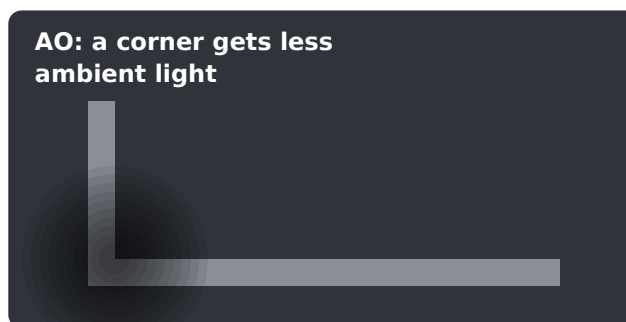
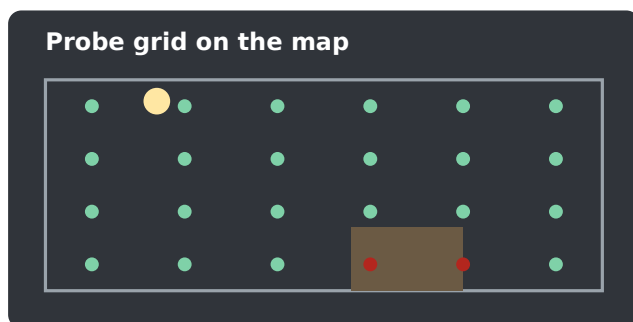
Stock openQ4 SSAO: noise and grey blotches on flat walls

GTAO from this pack: shading only where it really is

If something is wrong

What you see	What to do
Shadows are black again on some map	The map is not baked. Type r_lightGridDebug 1 : if no even grey haze covers the scene (black stays black), run bake_one_map.bat and pick that map.
The bake crashed with Out of memory	The paging file was not enlarged. Open the window from step 1 and check that the drive shows 131072 - 131072 (both numbers equal), not "None". Then run the .bat again - it skips finished maps.
The PC froze while baking	Only bake through our .bat files: they run the game in a window and at low priority.
The game now starts in a window	Console: seta r_fullscreen 1 , then vid_restart .
The lighting looks as if there is no preset	Check whether zzz_bake_helper.pk4 was left in baseoq4 after an interrupted bake, and delete it.
Nothing changed at all	1) Command path : is your zz_light_... listed? 2) Is OpenGL selected in the graphics settings, not Vulkan? 3) Did you type the exec command from step 4?
skipping missing map in the log	That's fine: the map list contains a name your game version doesn't have. The bake just moves on to the next map.
I want to remove everything	Delete zz_light_... from baseoq4 and type seta r_ssao 0 in the console. The baked data can stay - without a preset it changes nothing.

How it works (for the curious)



Baking. openQ4 places a grid of invisible "probes" over the map (green dots) and from each one takes a panoramic shot in every direction. That tells it how much light reaches that point from lit walls and floors. Each "bounce" is one more round: light reflected twice and three times reaches the most hidden corners. Probes that happen to end up inside objects (red) are ignored by the pack's shaders - otherwise those objects would look overexposed.

AO (ambient occlusion). A corner, gap or niche receives ambient light from only part of the directions, so it is darker. The shader computes this every frame from what is on screen. In preset A only the bounce light is darkened - lamp light and its shadows stay as designed.

The pack consists of modified openQ4 shaders and is distributed under the GNU GPL v3 (folder 4_SOURCE). No Quake 4 game data is included.